

Loan Approval Prediction Using Machine Learning

Introduction

Loan approval prediction is an important application of Machine Learning in the banking sector. Banks receive thousands of loan applications every year and need to decide whether an applicant is eligible for a loan. Making this decision manually can be time-consuming and may lead to errors. Machine learning helps automate this process by analyzing applicant information and predicting whether a loan should be approved.

The objective of this project is to build a machine learning model that predicts the loan approval status based on applicant details such as income, education, employment status, loan amount, loan term, CIBIL score, and asset values.

Dataset Description

The dataset used in this project contains **4,269 records** and **13 features**. Each record represents one loan applicant.

The dataset includes the following features:

- Loan ID
- Number of Dependents
- Education
- Self Employed Status
- Annual Income
- Loan Amount
- Loan Term
- CIBIL Score
- Residential Assets Value
- Commercial Assets Value
- Luxury Assets Value
- Bank Asset Value
- Loan Status (Target Variable)

During data understanding, the dataset shape, column names, data types, missing values, and duplicate records were checked. The dataset contained no significant missing values, and duplicate records were removed if present.

Data Preprocessing

Before training machine learning models, the dataset was preprocessed.

The following preprocessing steps were performed:

- Checked for missing values.
- Removed duplicate records.
- Converted categorical variables (Education, Self Employed, and Loan Status) into numerical values using Label Encoding.
- Standardized numerical features using StandardScaler to improve model performance.

These preprocessing steps ensured that the dataset was clean and suitable for machine learning algorithms.

Exploratory Data Analysis (EDA)

Several visualizations were created to understand the data.

The following graphs were generated:

- Loan Status Analysis
- Education Distribution
- Income Distribution
- Income vs Loan Approval
- CIBIL Score Distribution
- Correlation Heatmap

The visualizations helped identify relationships among variables and understand the characteristics of loan applicants.

Machine Learning Models

Three classification algorithms were trained for loan approval prediction:

1. Logistic Regression

Logistic Regression is a simple and widely used classification algorithm that predicts the probability of loan approval.

2. Decision Tree Classifier

Decision Tree creates a tree-like structure using decision rules based on different features of the dataset.

3. Random Forest Classifier

Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Model Evaluation

The performance of each model was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These evaluation metrics provide a detailed comparison of model performance and prediction quality.

Results

After training and evaluating the three models, their performances were compared. Random Forest generally achieved the highest accuracy and F1-score, making it the best-performing model for this dataset. Logistic Regression also performed well, while the Decision Tree model showed comparatively lower performance.

Conclusion

This project successfully developed a machine learning model for predicting loan approval. Data preprocessing and exploratory data analysis improved the quality of the dataset before training the models.

Among the three algorithms, the Random Forest Classifier performed the best based on evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. Therefore, Random Forest is recommended as the most suitable model for loan approval prediction because it provides more accurate and reliable results.